HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed mechanical components and assemblies for new machine development projects, creating **3D CAD models** and **Bills of Materials**.
- Supported new machine projects from concept through prototype build, testing, and release to production, applying **Design for Manufacture and Assembly** principles.
- Investigated production and field issues to identify root causes and implement **corrective actions**, improving existing machine reliability and manufacturability.
- Developed and maintained accurate **3D CAD models** and **manufacturing drawings** for mechanical assemblies, ensuring adherence to design specifications.
- Collaborated with production, manufacturing, and purchasing departments to integrate new designs and improve existing machines for cost and performance.
- Applied knowledge of **tolerances, fits, GD&T;, materials, and manufacturing processes**, including sheet metal and machining, to component design.
- Containerized the feature-pipeline services with **Docker** and deployed them to a Kubernetes cluster (EKS), wiring up Helm chart configuration and basic resource limits for predictable rollout.

SKILLS

CAD & Design – 3D CAD modelling, Creo, SolidWorks, Inventor, CATIA, Siemens NX, GD&T;, DFM/DFA, BOM Management

Engineering Principles – Mechanical Design, Manufacturing Engineering, Product Design, Agricultural Engineering, Problem-Solving, Continuous Improvement, FMEA, Testing & Validation

Software & Tools – Creo, SolidWorks, Inventor, CATIA, Siemens NX, PLM/PDM, Jira, Confluence, Git, Jenkins

Manufacturing Processes – Sheet Metal, Weldments, Machining, Castings, Plastics, Hydraulics, Electrical Integration, Mobile Machinery

General Skills – Communication, Teamwork, Attention to Detail, Work Ethic, Adaptability, Resourcefulness

PROJECTS

Agricultural Machine Component Design Optimization

- Designed and modelled a new hydraulic manifold assembly for a prototype agricultural implement using **SolidWorks**, focusing on **reducing part count** by 18% through part consolidation.
- Produced detailed **2D manufacturing drawings** with GD&T; annotations for a custom gearbox housing, enabling precise machining and ensuring **interchangeability** with existing components.
- Collaborated with the workshop team to review prototype builds, identifying potential assembly issues and refining the design for improved **manufacturability**.

Automated Production Layout and Jigs (HCL Technologies)

- Developed Python scripts to automate the generation of initial 2D production line layouts based on Bill of Materials data, reducing manual layout design time by **~40%**.
- Designed and documented requirements for custom jigs and fixtures in Java Spring Boot applications to support manufacturing process automation and improve assembly accuracy for enterprise customers.
- Assisted in the integration of these automated tools into the existing CI/CD pipeline using Jenkins, ensuring consistent application of design standards across multiple projects.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)